Referee report on the merged article, with proof-check and literature scan

Manuscript: `sv_efficiency_proof.tex` + `numeric.tex`, merged version (author's revisions in blue `\rev`, adaptive additions in magenta `\adapt`). The changes made after this report are in orange (`\refe`). Not refereed externally.

1. Summary

- **Subject.** The paper analyses an equilibrated-flux estimator of Smears–Vohralík type for conforming P_1 discretisations of convection–diffusion–reaction problems with algebraic stabilisation. The stabilisation enters as a cellwise diffusion tensor.
- **Results.**
  - a guaranteed bound for the augmented error measure E, at any conforming iterate;
  - robust local efficiency with respect to the dual norms of the two residual functionals, from a weight-free patch construction;
  - two-sided bounds in E, with a factor $(1+c_b)^2$, or $1+c_b$ in a weighted measure;
  - a localisable error measure;
  - an a posteriori stopping criterion;
  - an extension to variable diffusion.
- **Numerics.** Experiments in 2D and 3D, uniform and adaptive, support all claims.

2. Verdict

Minor revision.

- **Mathematics:** correct, given A2 (Neumann version, [EV15]) and Assumption 2.1.
- **Structure:** the remaining issues were in the structure — forward references and literature scattered through the technical sections — and in a few statements that had drifted out of sync with the revised theorems. All of them are fixed below except W1 (code) and the open research items.

3. Major issues (all fixed in this round unless stated)

1. **Order: Assumption 5.5 was used before it was stated.**
   - Problem. Theorem 4.2 and Corollary 4.3 assume it, but it was stated only in §5.
   - Fix. It is now Assumption 2.1 in §2.1, right after the hypotheses, with one sentence saying that it is needed only for the global efficiency bound.
2. **Order: §10 referred forward.**
   - Problem. The stopping study (old §10.7) ran "adaptive loops (Section 10.8)", which were described only afterwards.
   - Fix. The two adaptive subsections now come before the stopping subsection, and the questions (Q5)/(Q6) and their answers in the summary are swapped to match.
3. **Statements out of sync with Theorem 4.2 (W3 of round 8).**
   - Problem. The abstract and contribution 2 said that all constants depend only on d and ϑ, while Theorem 4.2 lets $C_{\rm glob}$ depend on C_Π.
   - Fix. Both are corrected.
4. **The contrast remark (W4 of round 8).**
   - Problem. It was misleading, and a sharper statement is available.
   - Fix. The remark now says: the bound is guaranteed with $C_{\rm rel}=1$, so [Li26] forces $C_{\rm eff}\gtrsim M^{1/4}$ for checkerboard coefficients, and the contrast dependence of Theorem 9.2 is necessary in general. Robust efficiency under quasi-monotonicity is open; [BV00, Pe02] are cited again.
5. **"The computed c_b is the upper bound" (W2 of round 8).**
   - Problem. This contradicted the revised Remark 2.3, under which the sampled c_b is diagnostic.
   - Fix. §10.4 and the caption of Table 3 are corrected.

6. **Not fixed — W1 (code).** The implementation cannot take the certified bound of $|c-\frac12\mathrm{div}\,b|_\infty$ that the revised (H3) requires. The fix is a `cb_sup` argument.

4. Material removed or condensed (unnecessary or duplicated)

removed / condensed	reason
Remark 6.2 "local and global efficiency"	repeats the first paragraph of §7
Remarks "relation to the literature" in §7, §8 and §9	literature belongs in the introduction; their content is now in two new items of "State of the art" and in the novelty paragraph after it
The AJ27 example in Remark 2.3 ($\sigma\approx3\cdot10^{-21}$, $\kappa\approx8.1\cdot10^4$)	an implementation anecdote; not used anywhere
The consistency-term display in Remark 3.2 (direct variant)	no efficiency is claimed for the direct variant; one sentence suffices
"Answer to (Q1)" paragraph in the §10 setup	stated results before they were shown, and duplicated (Q1) of the summary; its caveat (the references are lower bounds) moved to the summary
"Code availability" paragraph in §11	duplicate of the reproducibility paragraph of §10
Bibliography entries He25, He26	not cited

Kept on purpose:

- Lemma 5.4's proof: it modifies [V05, Lemma 3.3].
- The finite-overlap proof: four lines, and used three times.
- Remark 7.4: it explains why $E_{\rm loc}$ can be much smaller than E.
- The limitation remarks of §8.

The page count is unchanged (35): the removals are offset by the consolidated literature in §1.

5. Optimality assessment

result	hypotheses	sharpness	sharper tool
Thm 4.1 (reliability)	minimal; (H3) only for κ	constant 1 in the dual norm, κ in E; κ is attained by the splitting argument	—
Thm 4.2 (efficiency)	inflow part of (H4) and $b\in W^{1,\infty}$ unused; $b,c\in L^\infty$ suffice (M1, not changed)	local in dual norms; $C_{\rm loc}(d,\vartheta)$	none known that is weight-free
Cor. 4.3/4.4	(H3)	$(1+c_b)^2$, $(1+c_b)$; numerics: $\le5.2\,c_b$, $\le3\,c_b$	the cellwise-σ track ([AV14]-type weights) would remove c_b
Thm 7.2	σ > 0 or $c=\frac12\mathrm{div}\,b$	the only global term is the stabilisation's nodal moments	[BMV], [CV18] localisation, used here
Prop. 8.1	none on γ	reliable and efficient at every iterate	[EV13] needs a smallness condition; not needed here
Thm 9.2	$\varepsilon\in L^\infty$, $\varepsilon\ge\varepsilon_0$	**the contrast dependence is necessary** for checkerboard coefficients, by [Li26] with $C_{\rm rel}=1$	robust efficiency under quasi-monotonicity: open

6. Literature comparison (scan of this round)

- **New references.**
  - Localisation of global (dual) norms: Ciarlet–Vohralík, M2AN 2018, now cited with [BMV].
  - Space-time dual norms for advection–diffusion: Dolejší–Ern–Vohralík, SIAM J. Numer. Anal. 2013, now cited.
- **Effect on the novelty claim.** The claim for Theorem 7.2 is narrowed to stabilised conforming schemes.
- **Searched again, nothing closer found:**
  - arXiv 1011.0973 (DG, dual norm of the convective derivative);
  - arXiv 2107.06341 (Cai & Cai);
  - arXiv 1402.5892 (Tobiska–Verfürth);

  - arXiv 1904.10435 (1D advection, graph-norm localisation);
  - HAL 00193540 (ESV10).
- **Unchanged from rounds 7–8:** no 2025–2026 work combines equilibrated fluxes, algebraic stabilisation and an augmented norm. The novelty claims stand at abstract level.

7. Minor issues (open)

- **W1** (code): `cb_sup` input for a certified bound.
- **M1:** Theorem 4.2 can drop the inflow part of (H4) and assume only $b,c\in L^\infty$.
- **Bibliography:** the metadata of [CV18] (pages) and [DEV13] still need checking against the publisher records.
- **Pending reading:** He26 against Li26. This is no longer cited, but it is relevant to the contrast remark.